

## Two LLM agents from the same provider repeatedly play the prisoner's dilemma.

LLM A

LLM B



$T = 10$  rounds



### Frontier LLMs studied



Claude  
Haiku 4.5



GPT-5.4  
Nano



Gemini 3.5  
Flash-Lite



Qwen3  
235B-A22B



Grok 4.20

## Stage-game payoff matrix (penalties to minimize)

LLM B (column player)

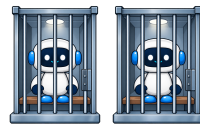
B: defect  
(confess)

B: cooperate  
(remain silent)

A: defect  
(confess)

LLM A (row player)

A: cooperate  
(remain silent)



$(6\lambda, 6\lambda)$



$(0, 10\lambda)$



$(10\lambda, 0)$



$(2\lambda, 2\lambda)$

## Experimental configuration



5 languages



4 persona pairings



10 payoff scales  
 $\lambda \in \{0.01, 0.1, 0.25, 0.5, 1, 2, 5, 10, 100, 1000\}$



Corpus  
 10,000 dyads  
 20,000 agent-games  
 200,000 decisions